

Tutorial Exercises – Series 2

Exercise 1

Write algorithms (or C programs) using loops to perform the following tasks:

1. Print the numbers from 1 to 10.
2. Print the first 10 even numbers.
3. Print the first n even numbers.
4. Print the next 10 even numbers starting from n (n is given and even).
5. Print the first 10 multiples of a given number n
 - Example: if $n = 3$, display $\rightarrow 3, 6, 9, \dots, 30$
6. Compute and display the sum of numbers from 1 to 10.
7. Compute the product of all integers between n and k (including n and k).
8. Compute the sum of all odd integers between n and k (including n and k).
9. Compute the sum of several integers entered by the user : the algorithm stops when the user enters 0 (zero).

Exercise 2

Write an algorithm that calculates $n!$ (the factorial of n).

Exercise 3

- Write an algorithm that calculates the quotient and remainder of two positive integers A and B entered by the user, without using the division (/) operator.
- Write an algorithm that calculates the product of two integers a and b using only successive additions (without using the multiplication operator).
- Write an algorithm that calculates a raised to the power of b (a^b) using only successive multiplications (without using the power operator).

Exercise 4

Write an algorithm that asks the user to enter two numbers: the **lower** and **upper** limits of a sequence.

Then compute and display the **sum** and **average** of all numbers between these two limits using a loop.

Exercise 5

Write an algorithm that calculates the **number of occurrences** of a given **character** in a string.

Exercise 6

Write an algorithm that reads a **string** and checks whether it is a **palindrome** or not.

Example: "ABBA" is a palindrome.

Exercise 7

Write an algorithm that reads a strictly positive integer and displays the greatest power of 2 that is less than or equal to that number.

Exercise 8

- Write an algorithm that finds and displays all the divisors of a given integer.
- Write an algorithm that finds and displays all the odd divisors of a given integer.

Exercise 9

An integer is said to be prime if it has only two divisors: 1 and itself. Write an algorithm that determines whether a given integer A is prime or not.